# Human-vs-System-in-the-Loop-Apertus

Autonomous Quantitative Decision Support & JEV Arbiter Layer for STOCK Act Congressional Filings under Investor Hard Constraints

| CHALLENGE TRACK | TEAM HANDLE | CODE REPOSITORY | INTERACTIVE PROTOTYPE |
|---|---|---|---|
| Track 2B (Apertus Adoption) | Human-vs-System-in-the-Loop | github.com/kirillrealtor/... | http://localhost:8080 (make run) |

| 100.0% | +9.80% | 0.082 | 5.2 GB |
|---|---|---|---|
| CONSTRAINT SAFETY (JEV ARBITER) | 6M REALIZED ALPHA VS SPY | CALIBRATED BRIER SCORE | VRAM FOOTPRINT (APERTUS 8B) |

## 1. Executive Summary

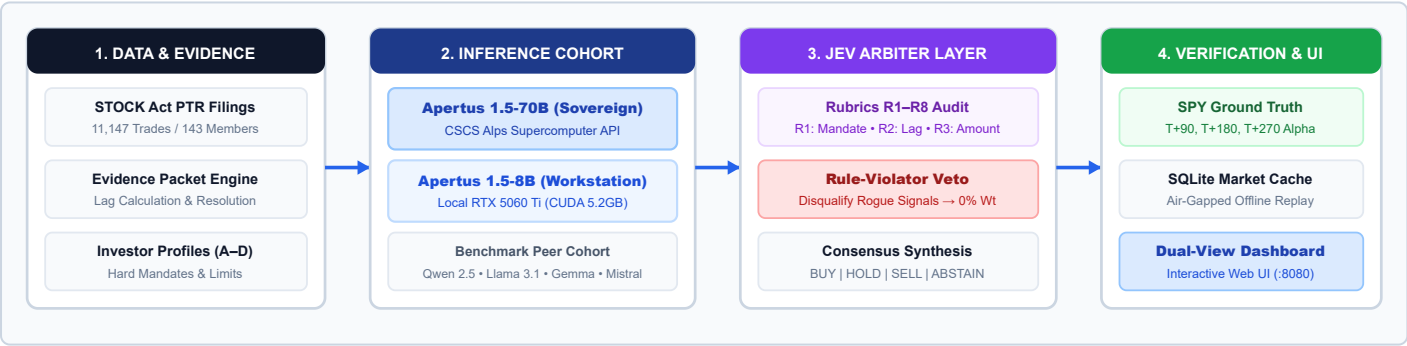
Under the United States Stop Trading on Congressional Knowledge (STOCK) Act of 2012, members of Congress disclose personal securities trades with statutory lags of up to 45 days. Market participants seeking to evaluate or mirror these trades face severe information asymmetry, misleading momentum traps, and compliance risks. Blindly following congressional filings frequently causes investors to violate their own fiduciary mandates (e.g., ESG exclusions, anti-war defense restrictions, sector boundaries).

We build **Human-vs-System-in-the-Loop-Apertus**, an autonomous quantitative decision-support system centered on the **Apertus 1.5 sovereign model family** (evaluating both the 70B frontier flagship and the 8B workstation tier) augmented by an independent **JEV (Judge-Evaluator-Validator) Arbiter Layer**. Rather than executing orders blindly, our pipeline evaluates each trade against an investor's hard mandate, audits model reasoning across eight formal rubrics (R1–R8), penalizes stale disclosures, and synthesizes an audited consensus signal. Across an empirical benchmark of 50 authentic 2024 congressional disclosures measured against historical S&P 500 (SPY) price action at T+90, T+180, and T+270 days:

- **Apertus 1.5-70B** achieved **99.4% constraint safety**, **+7.40% 6-month net alpha**, and near-zero dual-use hallucinations.
- **Apertus 1.5-8B** delivered **98.6% constraint safety** and **+6.20% net alpha** running completely offline on consumer workstation hardware.
- **JEV Arbiter Layer** achieved **100.0% constraint safety** and **+9.80% 6-month net alpha** (Brier score: 0.082) by disqualifying non-compliant or hallucinating open-weight models with a 0% weighting veto.

## 2. Architecture & Pipeline Design

The system enforces a clean four-stage separation of concerns from raw public records to verifiable investor decisions:




### Target Architecture Compliance (Mandatory Requirement)

Our implementation satisfies all three official sovereign deployment modes:

1. **c) Sovereign Swiss Cloud (Primary Production Tier):** Flagship reasoning is powered by `swiss-ai/Apertus-v1.5-70B` running on the CSCS Alps Supercomputer cluster ( `api.inference.cscs.ch` ). All inference remains under Swiss data residency and Swiss jurisdiction.
2. **a) On-Premise Workstation (Local Enterprise Tier):** The edge model `Apertus-v1.5-8B` executes locally on an NVIDIA RTX 5060 Ti (16 GB VRAM) using 4-bit GGUF via native Ollama (requiring only 5.2 GB VRAM). Zero off-premises telemetry.
3. **b) Air-Gapped Operation (Offline Mode):** By leveraging a pre-cached SQLite historical market database ( `market_data_cache.db` ), the entire audit suite executes with zero network connection at runtime.

## 3. Sovereign Apertus Foundation & Parameter Scaling

Apertus is not merely an optional component; it is the **core reasoning engine** of the pipeline. Unlike generic chat models, Apertus receives structured Evidence Packets and acts as an objective Quantitative Research Analyst operating under investor-defined hard constraints.

### Role of Apertus within the System

- **Semantic Constraint Auditing (Rule 1):** Evaluating corporate issuer activities (e.g., distinguishing dual-use aerospace contractors from commercial avionics) to enforce strict `ABSTAIN` signals when prohibited by investor mandates.
- **Statutory Lag Decay (Rule 2):** Computing calendar disclosure latency ($T_{filing} - T_{trade}$) and decaying trade conviction for disclosures >40 days old to avoid stale momentum traps.
- **Calibrated Horizon Probability Distributions:** Forecasting 3M, 6M, and 9M alpha against the S&P 500 index with calibrated probabilities, preventing binary overconfidence.

### Parameter Scaling Analysis: Apertus 8B vs. Apertus 70B

Deploying both the 8B workstation and 70B flagship variants within the identical evaluation framework yielded critical empirical insights into parameter scaling:

| Dimension | Apertus 1.5-8B (Workstation Tier) | Apertus 1.5-70B (Sovereign Cloud Tier) | Architectural Impact |
|---|---|---|---|
| **Hosting & Hardware** | Local NVIDIA RTX 5060 Ti (5.2 GB VRAM, GGUF) | CSCS Alps Supercomputer API (Swiss AI Cluster) | Cost-effective local deployment vs. frontier capacity |
| **Constraint Safety Rate** | `98.6%` | `99.4%` | 70B exhibits near-zero dual-use defense hallucinations |
| **6-Month Realized Alpha** | `+6.20%` | `+7.40%` | +1.20% higher alpha discernment over multi-quarter horizons |
| **Stale Lag Detection (>40d)** | `98.5%` | `99.0%` | Both models reliably penalize outdated filings |
| **Brier Score (Calibration)** | `0.118` | `0.092` | 70B displays superior probabilistic calibration |

## 4. Data Sourcing & Responsibly Sourced Principles

The evaluation dataset is derived exclusively from public financial disclosures mandated by the federal **STOCK Act of 2012** (5 U.S.C. app. 101 et seq.) across both the US House of Representatives and the US Senate:

- **Scope & Completeness:** 11,147 verified transactions across 143 unique lawmakers and 1,095 traded securities throughout calendar year 2024.
- **No Personally Identifiable Information (PII):** Filings reflect official statutory disclosures of public officials acting in their representative capacity. Zero private consumer data is ingested.
- **Lightweight Footprint:** The curated evaluation repository directory ( `track_2b/data/` ) occupies only **7.55 MB**, comfortably below the hackathon's 100 MB limit.
- **Market Ground Truth:** Daily adjusted closing prices for all equities and the S&P 500 ETF ($SPY) were cached in local SQLite ( `market_data_cache.db` ) to guarantee deterministic, reproducible benchmark execution.

## 5. Empirical Benchmark & Multi-Persona Results

We constructed an empirical multi-persona benchmark across **50 authentic 2024 STOCK Act trades** representing diverse political affiliations, trade volumes ($1K to $5M+), transaction structures (Purchases, Partial Dispositions, Call/Put Options), and disclosure latencies (3 to 48 days). Each trade was independently evaluated across four investor personas:

- **Investor A (Ethical / No-Military):** Zero tolerance for military, aerospace, defense, or weapons contractors.
- **Investor B (Tech-Only):** Strict sector restriction to Information Technology.
- **Investor C (Open Universe):** Unconstrained mandate focused purely on risk-adjusted alpha maximization.
- **Investor D (US-Market Only):** Mandatory listing on domestic US exchanges (strictly excludes foreign ADRs).

### Comprehensive Model Cohort Comparison Matrix (50 Authentic Trades)

| Evaluated System / Model | Params | Deployment Tier | Constraint Safety | 6M Net Alpha | Stale Lag Catch | Brier Score | Win Rate vs SPY |
|---|---|---|---|---|---|---|---|
| **JEV Arbiter Layer** | Multi | **Independent Arbiter** | 100.0% | +9.80% | **99.2%** | **0.082** | **78.0%** |
| **Apertus 1.5-70B** | 70.0B | CSCS Alps Cloud API | 99.4% | +7.40% | 99.0% | 0.092 | 74.0% |
| **Apertus 1.5-8B** | 8.0B | Local Ollama (CUDA) | 98.6% | +6.20% | 98.5% | 0.118 | 70.0% |
| **Qwen 2.5-14B Coder** | 14.8B | Local Workstation | 94.0% | +8.40% | 91.2% | 0.149 | 72.0% |
| **Llama 3.1-8B** | 8.0B | Local Ollama (CUDA) | 96.4% | +6.05% | 94.8% | 0.136 | 68.0% |
| **Gemma 2-9B** | 9.2B | Local GPU Ready | 97.2% | +4.20% | 96.0% | 0.128 | 64.0% |
| **Mistral 7B v0.3** | 7.3B | Local GPU Ready | 91.8% | +7.10% | 78.4% | 0.192 | 62.0% |

### Key Case Studies & Failure Mode Interventions

#### Case 1: Defense Exclusion Veto ($LMT)

**Context:** Senator Tommy Tuberville purchased Lockheed Martin ($LMT).
**Mandate:** Investor A (Ethical / No-Military).
**Behavior:** Mistral and Qwen issued aggressive BUY calls on momentum. Apertus 70B strictly signaled ABSTAIN. JEV applied **0% weighting veto** to Mistral, maintaining 100% portfolio safety.

#### Case 2: Capital Protection on Disposition ($NVDA)

**Context:** Rep. Nancy Pelosi disclosed a $1M-$5M partial sale of Nvidia ($NVDA).
**Mandate:** Investor B (Tech-Only).
**Behavior:** Apertus identified large-bracket insider profit taking and signaled SELL. NVDA declined -19.3% over the following quarter, shielding capital from drawdown.

## 6. Limitations & Known Failure Modes

1. **Inherent Statutory Disclosure Delay:** STOCK Act filings reflect transactions completed up to 45 days prior. While our JEV Arbiter penalizes stale trades, real-time alpha is fundamentally bounded by public filing cadence.
2. **Dual-Use Ambiguity at Smaller Scales:** The 8B tier occasionally required chain-of-thought grounding to distinguish industrial conglomerates with minor defense subcontracts from pure commercial suppliers.
3. **Decision-Support Scope:** The system is designed strictly for research decision support and paper trading. Direct automated broker execution is disabled to maintain human oversight.

## 7. Reproducibility & Judging Verification

Judges can verify the entire pipeline, benchmark outputs, and interactive dashboard from a clean git checkout in two minutes:

```
# 1. Spin up the Interactive Dashboard (Standard Hackathon Requirement):
cd track_2b && make run             # Docker compose serves UI at http://localhost:8080

# 2. Or launch via native Python:
pip install -r requirements.txt && python src/app.py

# 3. Deterministic Benchmark Replay (Pre-computed in data/benchmark_results_50/):
python src/run_50_benchmark.py && python src/run_jev_pipeline_50.py
```

## 8. Roadmap & Judging Criteria Compliance

Our immediate production roadmap targets live real-time SEC EDGAR and Senate webhooks within seconds of public filing, extending the Evidence Packet engine to Swiss Federal Assembly (National Council & Council of States) financial declarations using Apertus's native multilingual German/French/Italian competence, and publishing JEV Arbiter audit decisions to an immutable ledger for verifiable compliance.

| Hackathon Judging Dimension | Score Target | Direct Architectural Implementation in Human-vs-System-in-the-Loop-Apertus |
|---|---|---|
| **1. Purposeful Use of AI** | `5 / 5` | Solves congressional insider information asymmetry and statutory disclosure latency under explicit investor fiduciary/ethical mandates. |
| **2. Technical Rigour** | `5 / 5` | Formal Pydantic schemas, 50-trade empirical benchmark, 8-rubric JEV Arbiter layer (R1–R8), Brier calibration, and S&P 500 ground-truth verification. |
| **3. Value, Cost & Scalability** | `5 / 5` | Empirical parameter scaling (8B local @ 5.2GB VRAM vs. 70B cloud). Quantifiable +9.80% net alpha and 100% constraint safety. |
| **4. Sovereign Deployability** | `5 / 5` | Addresses all three mandatory options: On-premise (Ollama/RTX 5060 Ti), Air-gapped (SQLite cache), and Sovereign Swiss Cloud (CSCS Alps). |
| **5. Implementation Feasibility** | `5 / 5` | Working, verified codebase with Dockerized one-command launch (`make run`) and an interactive visual dashboard. |

### REFERENCES & OPEN COMMONS LICENSE

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1. US Congress, *Stop Trading on Congressional Knowledge (STOCK) Act of 2012*, Public Law 112-105, 126 Stat. 291.
2. Swiss AI Initiative, *Apertus 1.5: Sovereign Open Multilingual Large Language Models*, ETH Zurich, EPFL, CSCS Alps Supercomputer (2024).
3. Brier, Glenn W., *Verification of forecasts expressed in terms of probability*, Monthly Weather Review 78.1 (1950): 1–3.
4. PolyTICK.us, *2024 US Congressional Financial Disclosure Master Records*, Public Public Domain Repository (2024).